

MS4215: Lab 9

Logistic Regression: Model Fit and Classification

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Introduction

This lab uses the **Pima Indians Diabetes** dataset from the **MASS** package, drawn from a study of Pima Indian women in Arizona. The dataset contains 532 women with no missing values. The outcome is **type**, a binary indicator of diabetes diagnosis.

The variables are:

Variable	Description
npreg	Number of pregnancies
glu	Plasma glucose concentration (mg/dL)
bp	Diastolic blood pressure (mmHg)
skin	Triceps skin fold thickness (mm)
bmi	Body mass index (kg/m ²)
ped	Diabetes pedigree function (measure of genetic predisposition)
age	Age in years
type	Diabetes status (No / Yes)

The lab has two parts. **Part A** fits a logistic regression model and assesses its goodness of fit using pseudo R^2 measures, the Hosmer-Lemeshow test, and a calibration plot. **Part B** converts the model's predicted probabilities into classifications and evaluates performance using a confusion matrix, receiver operating characteristic (ROC) curve, and area under the curve (AUC).

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("MASS")
#install.packages("tidyverse")
#install.packages("DescTools")
#install.packages("ResourceSelection")
#install.packages("pROC")
library(MASS)
library(tidyverse)
library(DescTools)
library(ResourceSelection)
library(pROC)

pima <- rbind(Pima.tr, Pima.te)
```

Questions

Part A: Model fit and diagnostics

(i) Explore the data

Use `summary()` to inspect the dataset. How many patients were diagnosed with diabetes? What proportion of the sample has diabetes?

Then produce two boxplots using `geom_boxplot()`:

- Plasma glucose (`glu`) by diabetes status (`type`)
- BMI (`bmi`) by diabetes status (`type`)

Use `fill = type` in each plot. Comment on whether glucose and BMI appear to differ by diabetes status.

(ii) Logistic regression and pseudo R^2

Fit the following logistic regression model:

```
m1 <- glm(type ~ glu + bmi + age + ped, data = pima, family = binomial)
```

Use `summary(m1)` to inspect the output. Which predictors are statistically significant at the 5% level?

Use `exp(coef(m1))` and `round(exp(confint(m1)), 3)` to obtain adjusted odds ratios and 95% confidence intervals. Interpret the adjusted OR for `glu`: what does it mean for a 10 mg/dL increase in plasma glucose?

Compute pseudo R^2 measures using:

```
PseudoR2(m1, which = c("CoxSnell", "Nagelkerke", "McFadden"))
```

Interpret the McFadden value against the 0.2–0.4 guideline.

(iii) Hosmer-Lemeshow test and calibration plot

Run the Hosmer-Lemeshow goodness-of-fit test with `g = 10` groups:

```
hoslem.test(as.numeric(pima$type) - 1, fitted(m1), g = 10)
```

Note: `as.numeric(pima$type) - 1` converts the factor to 0 (No) and 1 (Yes).

- Report the test statistic, degrees of freedom, and p-value.
- What is the conclusion?

Then build a calibration plot. Use `ntile(pred, 10)` to create deciles, `group_by()` and `summarise()` to compute the mean predicted probability and observed diabetes rate in each decile, and `ggplot()` with `geom_abline(intercept = 0, slope = 1, linetype = "dashed")` as the reference line. Comment on whether the model is well-calibrated.

Part B: Classification

(iv) Confusion matrix at $p = 0.5$

Compute the predicted probabilities from `m1`:

```
p_hat <- predict(m1, type = "response")
```

Use `ifelse(p_hat > 0.5, "Yes", "No")` to classify each patient and build a confusion matrix with `table()`.

From the table, compute:

- Sensitivity = $TP / (TP + FN)$
- Specificity = $TN / (TN + FP)$
- Accuracy = $(TP + TN) / n$

What accuracy would a null classifier achieve (one that always predicts “No diabetes”)? Is the model’s accuracy meaningfully better? Comment on whether sensitivity at $p = 0.5$ is adequate for a medical screening test.

(v) ROC curve and AUC

Fit a ROC object using `roc(pima$type, p_hat)` from the `pROC` package.

Plot the ROC curve using `ggplot()` with:

- `FPR = 1 - roc_obj$specificities` on the x-axis
- `TPR = roc_obj$sensitivities` on the y-axis
- A dashed 45° reference line

Compute the AUC using `auc(roc_obj)`. Classify it as poor, acceptable, excellent, or outstanding using the Hosmer-Lemeshow benchmarks (0.7–0.8 acceptable; 0.8–0.9 excellent; > 0.9 outstanding). Interpret the AUC value in plain language.

(vi) Choosing a threshold

Use `coords()` to find the Youden-optimal threshold:

```
coords(roc_obj, "best", best.method = "youden",  
      ret = c("threshold", "sensitivity", "specificity"))
```

Classify patients using this threshold and compute sensitivity and specificity. Present a comparison table of both thresholds:

Threshold	Sensitivity	Specificity
$p = 0.5$	...	...
Youden	...	...

In a diabetes screening context, missing a diabetic patient (a false negative) carries greater clinical risk than an unnecessary follow-up appointment (a false positive). Given this, which threshold would you recommend, and why?